

# Операционные системы

Основы интерфейса взаимодействия пользователя с системой Unix на уровне  
командной строки

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

# 1. Цели и задачи работы

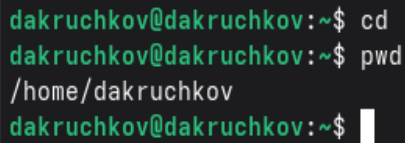
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Приобретение практических навыков взаимодействия пользователя с системой посредством командной строки.

- 1 Определить имя и путь домашнего каталога
- 2 Изучить команду `ls`.
- 3 Выполнить действия с каталогами.
- 4 Получить дополнительные сведения при помощи справки по командам.
- 5 Изучить команду `history`.

## 2. Процесс выполнения лабораторной работы

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A terminal window with a dark background and green text. It shows a user named 'dakruchkov' at a shell prompt '~\$'. The user enters 'cd' and then 'pwd'. The output of 'pwd' is '/home/dakruchkov'. The prompt is followed by a white cursor.

```
dakruchkov@dakruchkov:~$ cd
dakruchkov@dakruchkov:~$ pwd
/home/dakruchkov
dakruchkov@dakruchkov:~$
```

Рисунок 1: Путь к домашнему каталогу

```
dakruchkov@dakruchkov:~$ cd /tmp
dakruchkov@dakruchkov:/tmp$ ls
dd567469-1d41-4d55-ba5e-27e41626447d.zip
happd.sock
node-compile-cache
par-64616b727563686b6f76
snap-private-tmp
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-abrt.service-ONPREu
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-chrond.service-rL3gTK
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-colord.service-BkNiXY
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-dbus-broker.service-t4tj79
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-fwupd.service-b6itJB
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-irqbalance.service-61Hx7A
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-low-memory-monitor.service-uSweM
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-ModemManager.service-ItaFuG
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-pcsd.service-1h8x0R
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-polkit.service-rX6fqv
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-rtkit-daemon.service-G8kgWJ
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-switcheroo-control.service-W0cIdc
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-systemd-logind.service-ECMOI0
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-systemd-resolved.service-ZVEgPF
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-upower.service-mMPE8Z
VMwareDnD
vmware-root_895-3979642976
dakruchkov@dakruchkov:/tmp$
```

Рисунок 2: Команда ls



```
dakruchkov@dakruchkov:/tmp$ ls -la
.
..
dd567469-1d41-4d55-ba5e-27e41626447d.zip
.font-unix
happd.sock
.ICE-unix
node-compile-cache
par-64616b727563686b6f76
snap-private-tmp
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-abrt.service-ONPREu
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-chrond.service-rL3gTK
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-colord.service-BkNiXY
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-dbus-broker.service-t4tj79
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-fwupd.service-b6itJB
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-irqbalance.service-61Hx7A
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-low-memory-monitor.service-uSwhem
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-ModemManager.service-ltaFuG
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-pcsd.service-1hBx0R
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-polkit.service-rX6fqv
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-rtkit-daemon.service-G8kgWJ
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-switcheroo-control.service-W0cIdc
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-systemd-logind.service-ECMOIO
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-systemd-resolved.service-ZVEgPF
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-upower.service-mMPE8Z
VMwareDnD
vmware-root_895-3979642976
.X0-lock
.X11-unix
.X1-lock
.XIM-unix
dakruchkov@dakruchkov:/tmp$
```

Рисунок 3: Команда ls -la

```
dakruchkov@dakruchkov:/tmp$ ls -l
total 3872
-rw-r--r--. 1 dakruchkov dakruchkov 3961995 Sep  5 11:19 dd567469-1d41-4d55-ba5e-27e41626447d.zip
-rw-rw-rw-. 1 root      root      0 Sep  5 11:18 happd.sock
drwxr-xr-x. 3 dakruchkov dakruchkov 60 Sep  5 11:41 node-compile-cache
drwx----- 3 dakruchkov dakruchkov 60 Sep  5 11:31 par-64616b727563686b6f76
drwx----- 2 root      root      40 Sep  5 11:17 snap-private-tmp
drwx----- 3 root      root      60 Sep  5 11:17 systemd-private-7d13fa7ae6c44eafba6cf51dd76f
b2cb-abrted.service-0NPREu
drwx----- 3 root      root      60 Sep  5 11:17 systemd-private-7d13fa7ae6c44eafba6cf51dd76f
b2cb-chronyd.service-rL3gTK
drwx----- 3 root      root      60 Sep  5 11:18 systemd-private-7d13fa7ae6c44eafba6cf51dd76f
b2cb-colord.service-BkNiXY
drwx----- 3 root      root      60 Sep  5 11:17 systemd-private-7d13fa7ae6c44eafba6cf51dd76f
b2cb-dbus-broker.service-t4tj79
drwx----- 3 root      root      60 Sep  5 11:18 systemd-private-7d13fa7ae6c44eafba6cf51dd76f
b2cb-fwupd.service-b6itJB
drwx----- 3 root      root      60 Sep  5 11:17 systemd-private-7d13fa7ae6c44eafba6cf51dd76f
b2cb-irqbalance.service-61Hx7A
drwx----- 3 root      root      60 Sep  5 11:17 systemd-private-7d13fa7ae6c44eafba6cf51dd76f
b2cb-low-memory-monitor.service-uSwhem
drwx----- 3 root      root      60 Sep  5 11:17 systemd-private-7d13fa7ae6c44eafba6cf51dd76f
b2cb-ModemManager.service-ltaFuG
drwx----- 3 root      root      60 Sep  5 11:18 systemd-private-7d13fa7ae6c44eafba6cf51dd76f
b2cb-pcsd.service-1hBx0R
drwx----- 3 root      root      60 Sep  5 11:17 systemd-private-7d13fa7ae6c44eafba6cf51dd76f
b2cb-polkit.service-rX6fqv
drwx----- 3 root      root      60 Sep  5 11:17 systemd-private-7d13fa7ae6c44eafba6cf51dd76f
b2cb-rtkit-daemon.service-68kgWJ
drwx----- 3 root      root      60 Sep  5 11:17 systemd-private-7d13fa7ae6c44eafba6cf51dd76f
b2cb-switcheroo-control.service-W0cIdc
drwx----- 3 root      root      60 Sep  5 11:17 systemd-private-7d13fa7ae6c44eafba6cf51dd76f
b2cb-systemd-logind.service-ECM0iO
drwx----- 3 root      root      60 Sep  5 11:17 systemd-private-7d13fa7ae6c44eafba6cf51dd76f
b2cb-systemd-resolved.service-ZVEgPF
drwx----- 3 root      root      60 Sep  5 11:17 systemd-private-7d13fa7ae6c44eafba6cf51dd76f
b2cb-upower.service-mMPeBZ
drwxrwxrwt. 2 dakruchkov dakruchkov 200 Sep  5 11:54 VMwareDnD
drwx----- 2 root      root      40 Sep  5 11:17 vmware-root_895-3979642976
dakruchkov@dakruchkov:/tmp$
```

Рисунок 4: Команда ls -l

```
dakruchkov@dakruchkov: /tmp$ ls -f
..
node-compile-cache
par-64616b727563686b6f76
VMwareDnD
dd567469-1d41-4d55-ba5e-27e41626447d.zip
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-fwupd.service-b6itJB
.X1-lock
.X0-lock
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-pcsd.service-1hBx0R
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-colord.service-BkNiXY
happd.sock
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-ModemManager.service-ItaFuG
vmware-root_895-3979642976
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-systemd-logind.service-ECMOI0
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-upower.service-mMPE8Z
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-switcheroo-control.service-W0cIdc
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-rtkit-daemon.service-G8kgWJ
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-polkit.service-rX6fqv
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-low-memory-monitor.service-uSwhEM
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-chrond.service-rL3gTK
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-irqbalance.service-61Hx7A
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-abrt.service-0NPREu
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-dbus-broker.service-t4tj79
systemd-private-7d13fa7ae6c44eafba6cf51dd76fb2cb-systemd-resolved.service-ZVEgPF
.font-unix
.XIM-unix
.ICE-unix
.X11-unix
snap-private-tmp
dakruchkov@dakruchkov: /tmp$
```

Рисунок 5: Команда ls -f

```
dakruchkov@dakruchkov: /tmp$  
dakruchkov@dakruchkov: /tmp$ cd /var/spool/  
dakruchkov@dakruchkov: /var/spool$ ls -l  
total 0  
drwxr-x--x. 1 root abrt 24476 Aug 31 13:37 abrt  
drwx-----. 1 abrt abrt 0 Dec 4 2025 abrt-upload  
drwxr-xr-x. 1 root root 66 Oct 23 2025 anacron  
drwx-----. 1 root root 18 Oct 23 2025 at  
drwx-----. 1 root root 0 Aug 6 2025 cron  
drwx--x---. 1 root lp 6 Dec 12 2025 cups  
drwxr-xr-x. 1 root root 0 Jul 30 2025 lpd  
drwxrwxr-x. 1 root mail 318 Sep 5 11:16 mail  
drwxr-xr-x. 1 root root 0 Jan 27 2026 plymouth  
dakruchkov@dakruchkov: /var/spool$
```

Рисунок 6: Каталог /var/spool

```
dakruchkov@dakruchkov:/var/spool$ cd
dakruchkov@dakruchkov:~$ ls
Desktop  Downloads  git-extended  Music  Pictures  Public  Templates  Videos  work
dakruchkov@dakruchkov:~$ ls -al
total 28
drwx----- 1 dakruchkov dakruchkov 370 Sep  5 11:53 .
drwxr-xr-x 1 root      root      318 Sep  5 11:16 ..
-rw----- 1 dakruchkov dakruchkov 4700 Sep  5 12:03 .bash_history
-rw-r--r-- 1 dakruchkov dakruchkov  18 Jul 23  2025 .bash_logout
-rw-r--r-- 1 dakruchkov dakruchkov 144 Jul 23  2025 .bash_profile
-rw-r--r-- 1 dakruchkov dakruchkov 680 Sep  5 11:41 .bashrc
drwx----- 1 dakruchkov dakruchkov 484 Sep  5 11:46 .cache
drwx----- 1 dakruchkov dakruchkov 338 Sep  5 11:46 .config
drwxr-xr-x 1 dakruchkov dakruchkov   0 Sep  5 11:18 Desktop
drwxr-xr-x 1 dakruchkov dakruchkov   0 Sep  5 11:18 Documents
drwxr-xr-x 1 dakruchkov dakruchkov   0 Sep  5 11:18 Downloads
-rw-r--r-- 1 dakruchkov dakruchkov 334 Nov 14  2025 .emacs
-rw-r--r-- 1 dakruchkov dakruchkov 235 Sep  5 11:53 .gitconfig
drwxr-xr-x 1 dakruchkov dakruchkov  74 Sep  5 11:50 git-extended
drwx----- 1 dakruchkov dakruchkov 136 Sep  5 11:24 .gnupg
drwx----- 1 dakruchkov dakruchkov  20 Sep  5 11:18 .local
drwxr-xr-x 1 dakruchkov dakruchkov  34 Oct 23  2025 .mozilla
drwxr-xr-x 1 dakruchkov dakruchkov   0 Sep  5 11:18 Music
drwxr-xr-x 1 dakruchkov dakruchkov   0 Sep  5 11:18 Pictures
drwxr-xr-x 1 dakruchkov dakruchkov   0 Sep  5 11:18 Public
drwx----- 1 dakruchkov dakruchkov 132 Sep  5 11:29 .ssh
drwxr-xr-x 1 dakruchkov dakruchkov   0 Sep  5 11:18 Templates
drwxr-xr-x 1 dakruchkov dakruchkov  18 Sep  5 11:31 .texlive2025
drwxr-xr-x 1 dakruchkov dakruchkov   0 Sep  5 11:18 Videos
drwxr-xr-x 1 dakruchkov dakruchkov  10 Sep  5 11:28 work
dakruchkov@dakruchkov:~$
```

Рисунок 7: Файлы в домашнем каталоге

```
dakruchkov@dakruchkov:~$  
dakruchkov@dakruchkov:~$ mkdir newdir  
dakruchkov@dakruchkov:~$ mkdir newdir  
mkdir: cannot create directory 'newdir': File exists  
dakruchkov@dakruchkov:~$ mkdir newdir/morefun  
dakruchkov@dakruchkov:~$  
mkdir letters memos misk  
dakruchkov@dakruchkov:~$ mkdir letters memos misk  
dakruchkov@dakruchkov:~$ ls  
Desktop  Downloads  letters  misk  newdir  Public  Videos  
Documents  git-extended  memos  Music  Pictures  Templates  work  
dakruchkov@dakruchkov:~$  
^[[200~rm letters/ memos/ misk/  
^[[201~dakruchkov@dakruchkov:~$ rm letters/ memos/ misk// misk/  
rm: cannot remove 'letters/': Is a directory  
rm: cannot remove 'memos/': Is a directory  
rm: cannot remove 'misk/': Is a directory  
dakruchkov@dakruchkov:~$  
rm -r letters/ memos/ misk/  
dakruchkov@dakruchkov:~$ rm -r letters/ memos/ misk/  
dakruchkov@dakruchkov:~$ rm -r newdir/  
dakruchkov@dakruchkov:~$ ls  
Desktop  Documents  Downloads  git-extended  Music  Pictures  Public  Templates  Videos  work  
dakruchkov@dakruchkov:~$
```

Рисунок 8: Действия с каталогами

```
./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_extension/
yamadharmaminted-quarto':
_extension.yml minted-quarto.lua

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/image':
solvay.jpg

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources':
cs1 tex

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/csl':
gost-r-7-0-5-2008-numeric.csl

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/tex':
preamble.tex

./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/scripts':
image-report mpv-shot
dakruchkov@dakruchkov:~$ ls -t
git-extended work Desktop Documents Downloads Music Pictures Public Templates Videos
dakruchkov@dakruchkov:~$
```

Рисунок 9: Команда ls -R и ls -t

```
dakruchkov@dakruchkov:~$ help cd
cd: cd [-L|[-P [-e]] [-@]] [dir]
    Change the shell working directory.

    Change the current directory to DIR.  The default DIR is the value of the
    HOME shell variable.  If DIR is "-", it is converted to $OLDPWD.

    The variable CDPATH defines the search path for the directory containing
    DIR.  Alternative directory names in CDPATH are separated by a colon (:).
    A null directory name is the same as the current directory.  If DIR begins
    with a slash (/), then CDPATH is not used.

    If the directory is not found, and the shell option `cdable_vars' is set,
    the word is assumed to be a variable name.  If that variable has a value,
    its value is used for DIR.

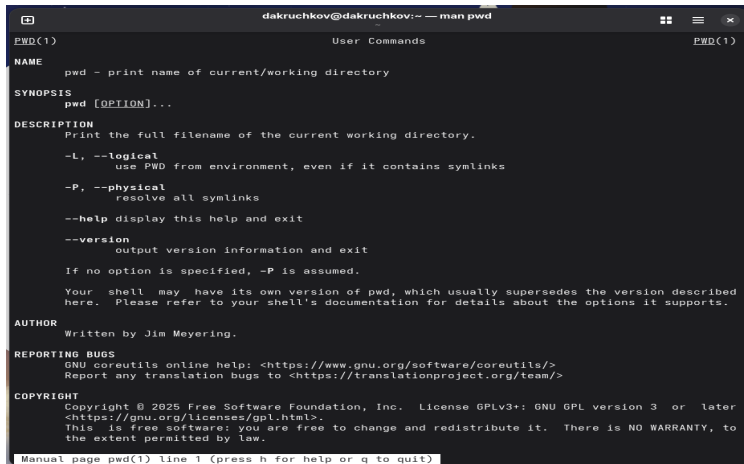
Options:
  -L      force symbolic links to be followed: resolve symbolic
          links in DIR after processing instances of `..'
  -P      use the physical directory structure without following
          symbolic links: resolve symbolic links in DIR before
          processing instances of `..'
  -e      if the -P option is supplied, and the current working
          directory cannot be determined successfully, exit with
          a non-zero status
  -@      on systems that support it, present a file with extended
          attributes as a directory containing the file attributes

The default is to follow symbolic links, as if `-L' were specified.
`..' is processed by removing the immediately previous pathname component
back to a slash or the beginning of DIR.

Exit Status:
Returns 0 if the directory is changed, and if $PWD is set successfully when
-P is used; non-zero otherwise.
dakruchkov@dakruchkov:~$
```

Рисунок 10: Справка по команде cd





```

dakruchkov@dakruchkov:~ — man pwd
PWD(1)                                User Commands                                PWD(1)

NAME
    pwd - print name of current/working directory

SYNOPSIS
    pwd [OPTION]...

DESCRIPTION
    Print the full filename of the current working directory.

    -L, --logical
        use PWD from environment, even if it contains symlinks

    -P, --physical
        resolve all symlinks

    --help
        display this help and exit

    --version
        output version information and exit

    If no option is specified, -P is assumed.

    Your shell may have its own version of pwd, which usually supersedes the version described
    here. Please refer to your shell's documentation for details about the options it supports.

AUTHOR
    Written by Jim Meyering.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>

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    the extent permitted by law.

Manual page pwd(1) line 1 (press h for help or q to quit)
```

Рисунок 11: Справка по команде pwd

```
MKDIR(1)                                User Commands                                MKDIR(1)

NAME
    mkdir - make directories

SYNOPSIS
    mkdir [OPTION]... DIRECTORY...

DESCRIPTION
    Create the DIRECTORY(ies), if they do not already exist.

    Mandatory arguments to long options are mandatory for short options too.

    -m, --mode=MODE
        set file mode (as in chmod), not a=rwx - umask

    -p, --parents
        no error if existing, make parent directories as needed, with their file modes unaffected by any -m option

    -v, --verbose
        print a message for each created directory

    -Z
        set SELinux security context of each created directory to the default type

    --context[=CTX]
        like -Z, or if CTX is specified then set the SELinux or SMACK security context to CTX

    --help
        display this help and exit

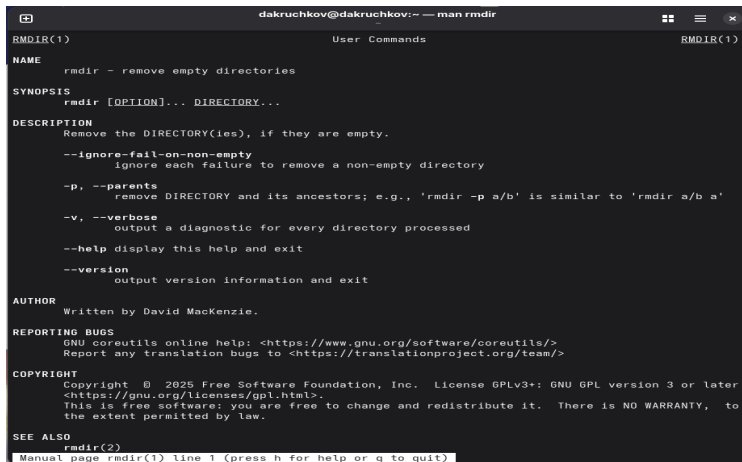
    --version
        output version information and exit

AUTHOR
    Written by David MacKenzie.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>

Manual page mkdir(1) line 1 (press h for help or q to quit)
```

Рисунок 12: Справка по команде mkdir



```
dakruchkov@dakruchkov:~ — man rmdir
RMDIR(1)                                User Commands                                RMDIR(1)

NAME
    rmdir - remove empty directories

SYNOPSIS
    rmdir [OPTION]... DIRECTORY...

DESCRIPTION
    Remove the DIRECTORY(ies), if they are empty.

    --ignore-fail-on-non-empty
        ignore each failure to remove a non-empty directory

    -p, --parents
        remove DIRECTORY and its ancestors; e.g., 'rmdir -p a/b' is similar to 'rmdir a/b a'

    -v, --verbose
        output a diagnostic for every directory processed

    --help display this help and exit

    --version
        output version information and exit

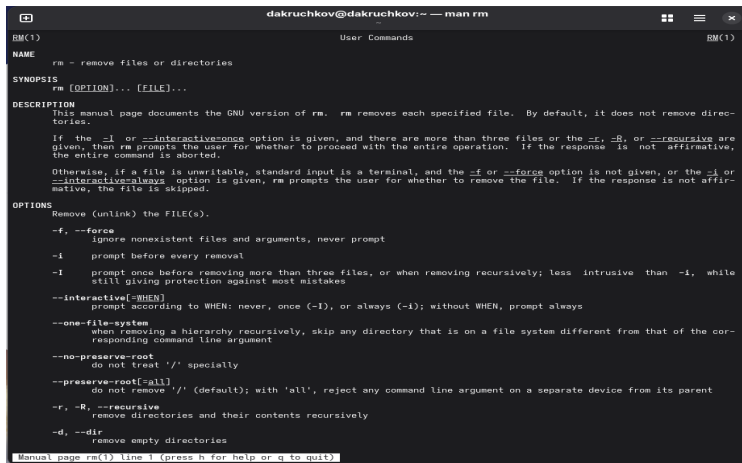
AUTHOR
    Written by David MacKenzie.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>

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    the extent permitted by law.

SEE ALSO
    rmdir(2)
    Manual page rmdir(1) line 1 (press h for help or q to quit)
```

Рисунок 13: Справка по команде rmdir



```
RM(1) User Commands RM(1)
NAME
rm - remove files or directories
SYNOPSIS
rm [OPTION]... [FILE]...
DESCRIPTION
This manual page documents the GNU version of rm. rm removes each specified file. By default, it does not remove directories.
If the -i or --interactive option is given, and there are more than three files or the -r, -R, or --recursive are given, then rm prompts the user for whether to proceed with the entire operation. If the response is not affirmative, the entire command is aborted.
Otherwise, if a file is unwritable, standard input is a terminal, and the -f or --force option is not given, or the -i or --interactive option is given, rm prompts the user for whether to remove the file. If the response is not affirmative, the file is skipped.
OPTIONS
Remove (unlink) the FILE(s).
-f, --force ignore nonexistent files and arguments, never prompt
-i prompt before every removal
-I prompt once before removing more than three files, or when removing recursively; less intrusive than -i, while still giving protection against most mistakes
--interactive[=WHEN] prompt according to WHEN: never, once (-I), or always (-i); without WHEN, prompt always
--one-file-system when removing a hierarchy recursively, skip any directory that is on a file system different from that of the corresponding command line argument
--no-preserve-root do not treat '/' specially
--preserve-root[=all] do not remove '/' (default); with 'all', reject any command line argument on a separate device from its parent
-r, -R, --recursive remove directories and their contents recursively
-d, --dir remove empty directories
Manual page rm(1) line 1 (press h for help or q to quit)
```

Рисунок 14: Справка по команде rm

```
215 pwd
216 cd /tmp
217 ls
218 ls -a
219 ls -l
220 ls -f
221 cd /var/spool/
222 ls -l
223 cd
224 ls
225 ls -al
226 mkdir newdir
227 mkdir newdir/morefun
228 mkdir letters memos misk
229 ls
230 rm letters/ memos/ misk/
231 rm -r letters/ memos/ misk/
232 rm -r newdir/
233 ls
234 ls -R
235 ls -t
236 help cd
237 man pwd
238 man mkdir
239 man rmdir
240 man rm
241 history
dakruchkov@dakruchkov:~$
```

Рисунок 15: Команда history

### 3. Выводы по проделанной работе

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Мы приобрели практические навыки взаимодействия пользователя с системой посредством командной строки.